

Neural HKR Claim-Field Validator

A Consolidated Machine-Learning Programme for the Empirical Test of
Holonomic Context Realism — Architecture, Falsification, and Scope

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Companion methodology note to the HKR monograph — a design programme, not an implementation.

Abstract. Three independent frontier models, asked how machine learning could support Holonomic Context Realism (HKR), converged on one design: a temporal graph model that learns HKR’s gauge-theoretic structure — holonomic curvature, two-channel (established / revision) dynamics, and redundancy — directly from the recorded history of science. This note consolidates those proposals and supplies what they omitted. It reframes the goal from *proving* HKR to *testing whether HKR’s operational structure has genuine predictive purchase on a real epistemic process*, beyond strong non-geometric baselines. It treats the scientific literature not as a watered-down proxy but as the theory’s natural home — a single field spanning the whole curvature axis, from contested claims to law-like ones — and it isolates the one boundary that remains: citation data tests the *epistemic mirror* of HKR, not its *physical referents*. The architecture is organised around falsification first: a confound gauntlet, an explicit treatment of the central tautology in the redundancy law, gauge identifiability, strict temporal-leakage discipline, and a claims ladder that mechanically bounds what each result licenses. The deliverable is the design and its rationale; no implementation is included, by intent.

Keywords: holonomic context realism; epistemic dynamics; temporal graph learning; gauge curvature; redundancy scaling; falsification; confound control; scientometrics.

0 The one decision that reframes everything

Before any architecture: a simulation — and a learned model — *cannot prove* a field theory of objectivity, of Quantum Darwinism in nature, or of bendable relativity. The HKR monograph says this in §13 (“a simulation establishes coherence and mathematical plausibility; it cannot, alone, prove a statement about nature”). Each of the three model threads quietly drifts back into the word *prove* (“you have proven HKR”, “the ultimate proof”, “validate the Revisionsaxiom in reality”). That drift is the single biggest threat to the project, because it invites exactly the dismissal a hostile reviewer wants to make. The mission is therefore restated precisely, and the whole architecture serves this one sentence and nothing grander:

Mission. Test whether HKR’s *operational structure* — a gauge-theoretic geometry of context transport, with curvature, redundancy-driven relaxation, and a two-channel (established / revision) dynamics — has **genuine predictive purchase on a real epistemic process** (the historical stabilisation of scientific claims), **beyond what strong non-geometric baselines already explain.**

If that holds under adversarial controls, the result is real and publishable: *HKR’s mathematics captures a measurable regularity in how knowledge becomes objective*. That is strong support for HKR *as a theory of epistemic dynamics*. Keeping it cleanly separated from HKR *as physics* (Section 2) is what makes the whole thing credible rather than grandiose.

1 Consolidation verdict — what survives from the three threads

The three conversations are three voices arriving, by different routes, at the same artifact (a “Neural HKR Claim-Field Validator with Differentiable Dynamics”). They are not three different ideas; they are one idea plus three different risk profiles. Table 1 records what each contributed and what survives.

Thread	Core proposal	Real contribution	Fatal flaw if taken alone
A (generative simulator)	Differentiable multi-agent active-inference extension of hkr_engine (PyTorch/JAX, HolonomyConv, Neural SDE, agents, holoeconomic bending)	The <i>mechanistic vocabulary</i> : how to make holonomy, curvature, free-energy flow, and the mutable core <i>differentiable</i> rather than Monte-Carlo.	It is synthetic. Running it harder only tightens <i>within-model</i> coherence — which you already have at 0.881. It never crosses the simulator→nature bridge that §13 says is the point. Cannot be the core.
B (T-GNN on citations)	Riemannian temporal GNN on real citation cascades to test P2 (percolation) and the $R^{-1/2}$ law	The <i>correct instinct</i> : external data, real epistemic history, falsifiable exponents.	Narrow (one prediction), and underspecified on confounds — no real defence against “ κ is just popularity”.
C (“third AI”)	Neural HKR Claim-Field Validator: validation-first, observable proxies (not “truth”), explicit success/failure, baselines, ablations, historical cutoffs	The <i>right backbone and the right epistemics</i> : predict observable stabilisation, never metaphysical truth; pre-commit to what would falsify.	Slightly too conservative — leaves the geometric machinery (the thing that makes it <i>HKR</i>) as an afterthought.
Synthesis (Gemini)	C-core + A/B differentiable dynamics + agents as capstone; ordering = validator → dynamics → agents	The <i>correct build order</i> , and the one genuine catch: do not hard-code $R^{-1/2}$ into the loss , or “discovering” it is worthless.	Still under-flags the deeper tautology (Section 6) and still blurs epistemic-HKR vs physics-HKR (Section 2).

Table 1: The three model threads as one idea seen through three risk profiles.

Consolidated decision. Keep **C** as the **spine** (falsification-first, observable proxies). Use **A/B’s differentiable geometry as an *inductive bias*, not a synthetic playground** — the holonomy/curvature machinery earns its place only if it improves prediction on real data. Demote **agents to a clearly-labelled exploratory capstone**. Then add the three things none of the threads supplied, which are the load-bearing parts of this note: (i) a clean *scope split* between epistemic-HKR and physics-HKR (Section 2); (ii) an explicit *confound gauntlet and circularity / CLT defence* (Sections 5–6) — without these the result is unfalsifiable and therefore not science; (iii) a *claims ladder* that mechanically prevents overreach while still reaching the radical core (Section 2.5).

2 What is actually being tested — one drifting field, law as its limit

2.1 No floor: the continuum from contested claim to physical law

An earlier draft drew a wall between “epistemic” and “physical” claims and put citation data on one side of it. That wall is not HKR, and removing it is the point of this section. HKR’s core thought is that there is **no view-from-nowhere floor**. There is one field of fact-states that drifts with context and time, and what we call an “objective fact” — even a “basic physical law” — is not a different *kind* of thing. It is the deepest, lowest-curvature, most-redundant region of that same field: the asymptote that claims relax *toward* as $\kappa \rightarrow 0$ and $R \rightarrow \infty$, and which in principle can itself drift. The theory is a unification along **one axis** (curvature down,

redundancy up), not a partition into two ontologies. So the ML programme does not amputate the radical core to stay safe — it is built to target it.

2.2 Why the literature is the theory’s natural home (not a watered-down proxy)

The scientific record is the ideal first domain precisely because it **spans the entire axis at once**. It contains everything from freshly-minted contested claims (high κ , low R) to claims so established they function as a field’s local physical floor ($\kappa \approx 0$, R enormous: conservation laws, the second law as used in practice, the Standard Model as a working assumption). HKR’s distinctive, checkable claim is that the gradient from *speculative* to *textbook-law-like* follows κ -relaxation **continuously** — that the most law-like items sit at the $\kappa \rightarrow 0$ / $R \rightarrow \infty$ end of *one smooth relationship*, not in a separate box. Demonstrating that **is** support for HKR’s unification claim itself: an effective law and a contested hypothesis as two regimes of a single field. That is the core thought tested, not denatured.

There is also an upgrade that strengthens the case rather than weakening it. In most domains O, E, T are lossy stand-ins for some hidden “real” quantity — which is what makes the word *proxy* feel like a concession. In the *epistemic* domain they are arguably *constitutive*, *not indicative*: for the practice of a field, the stabilised consensus **is** the local reality — there is no gap between “perceived as established” and “functions as established”. So O is not a proxy for an inaccessible truth; it is the actual HKR quantity, and citation data is the one place where manifestation and stabilisation are the *same act*. The literature is not where the theory is approximated — it is where the map is tightest.

2.3 The radical core, mapped to what the data can see

Each facet of “facts drift, and in drifting shift reality and perception” has a concrete, observable shadow in the literature (Table 2).

Facet of the HKR core thought	HKR construct	Testable epistemic shadow
Facts drift over context	Context-lability s_i (C16)	$\text{Var}_\theta[O]$ rises with κ — field-to-field / community-to-community acceptance variance, larger for high- κ cores.
Facts drift over time	Ontology-history revision (A4)	Stabilisation <i>and</i> de-stabilisation timelines (P4); status trajectories of claims.
Even the core / even basic laws drift	Mutable fact-core, false→true→false (C15)	Consensus reversals of once-foundational assumptions (simultaneity before/after relativity; parity, 1956; continental drift; the expanding universe; <i>H. pylori</i> vs the stress-ulcer dogma) — predicted to be the tail of one distribution , governed by the same κ -gated, alignment-signed, redundancy-relaxed dynamics, <i>not</i> a separate category.
Facts shift reality & perception	Mimesis / active inference (§6–§7)	Adoption <i>is</i> the mimetic stabilisation by which a claim becomes real-for-the-community — a mechanism to model, not a passive timeline.

Table 2: Each facet of the radical core has an observable counterpart in citation history.

The load-bearing, distinctive bet is the **continuity claim**: that foundational reversals are the extreme tail of the *same* relaxation process that governs ordinary claims, not a special exception. That is exactly what citation history can check, and exactly what would separate HKR from a tame “new ideas get popular” story.

2.4 The one genuine boundary — the mirror, not a retreat

Precision here is *faithfulness to HKR, not a smaller claim*. Citation data tests the **epistemic mirror**: how claims — at every regime, including the law-like end — stabilise and drift within human knowledge. It cannot test whether the **physical referents** themselves behave as drifting fact-fields: whether actual light cones deform under holoeconomic tension (P6), or dark-sector curvature leaks (P5). Those require physical measurement; *arXiv-about-physics is not physics-the-world*. Holding that line is what lets the credible epistemic result stand on its own instead of being dismissed alongside claims only a laboratory can adjudicate.

HKR already takes exactly this posture. §12 reads recent empirical results as “**resonance surfaces, not confirmations**”. By that standard, a strong positive epistemic result — the continuum from contested to law-like, the relaxation law on *independent* redundancy, the mutable-core reversals following one law — is precisely **resonance for the unification claim**. It is the strongest non-laboratory evidence the unification can receive, and it is the claim the theory itself says is the right one to make *before* the §13 lab programme.

2.5 The claims ladder (anti-overreach device)

Each tier is what a *result* licenses you to say; climb only when the rung below is secured under the controls in Section 5. The apex rung *reachable by this method* is the unification / continuity claim (rung 5). The genuine ceiling (rung 6) concerns the physical referents and is out of reach here by design.

Rung	What you may claim	What it takes
0	“HKR’s variables can be <i>operationalised</i> on real citation data.”	The proxies exist and are stable. (Cheap; not yet evidence.)
1	“An HKR-shaped model <i>predicts</i> future claim stabilisation.”	Beats a trivial baseline out-of-sample with leakage discipline.
2	“HKR structure predicts stabilisation beyond popularity, recency, prestige, novelty, and ordinary graph learning.”	Beats the full confound gauntlet (Section 5.1) across ≥ 2 domains. The minimum publishable headline.
3	“The learned curvature behaves as HKR specifically predicts — it relaxes with <i>independent</i> redundancy, and high- κ /high- T claims show distinct revision/labability dynamics.”	Section 6’s decoupled redundancy test + context-lability test survive ablation and shuffle.
4	“HKR’s <i>geometry</i> (holonomy / path-dependence), not just its features, is doing the work.”	Removing the loop/holonomy terms (parameter-matched) degrades prediction; context-shuffle destroys the signal.
5	“ Contested and law-like claims lie on one continuous κ/R axis — an effective law is the low-κ / high-R limit of the same field, and even foundational cores drift by the same dynamics. ”	The continuity-of-relaxation result across the full spectrum plus foundational-reversal cases following the same law. The radical core, tested — apex epistemic result.
6	“The physical <i>referents</i> themselves drift as fact-fields (light cones deform; P5/P6 in the world).”	Not reachable from citation data. Requires the §13 physical/observational programme. Reportable only as <i>resonance</i> , never confirmation.

Table 3: The claims ladder: each rung is what a result licenses, climbed only after the rung below is secured.

3 The scientific object: science modelled as an HKR fact-field

The model treats the literature as a discrete sampling of HKR’s continuous fact-field. The mapping is the part that makes this *HKR* rather than yet another citation-prediction paper, so it has to be exact.

Nodes = fact-states. The unit starts as a paper and graduates to a *claim-family* (a re-identifiable cluster across papers), because HKR’s *Faktkern* is precisely the re-identifiable core that survives transport, not the raw text. Each node carries the operational tuple $x_i(\theta, t) = (O_i(t), E_i(t), T_i(t), M_i(\theta, t))$ with fiber ψ_i .

The proxies (the soft underbelly — see Section 5.2). None is “truth”; all are observable stabilisation behaviour, and each must be anchored in something *not* trivially identical to the others:

- **E (evidential anchoring)** $\leftarrow$ independent replication-like uptake, benchmark reuse, method adoption; for theory fields, independent conceptual reuse.
- **O (ontic compatibility)** $\leftarrow$ centrality in the mainstream cluster, terminology stabilisation, decline of explicit-contradiction citations, textbook/review absorption. *Compatibility with the accepted ontology — not final truth*, exactly as HKR defines O .
- **T (transformation pressure)** $\leftarrow$ semantic novelty + cross-field disruption + downstream restructuring (an existing disruption index such as the CD / Funk–Owen measure is a natural anchor and, not coincidentally, a baseline you must beat).
- **R (redundancy)** $\leftarrow$ the **effective number of *independent* communities** carrying the claim. A diversity quantity (entropy / Simpson effective count over citing communities), *never* raw citation count, and — critically — **measured independently of the transport sample used to estimate κ** (Section 6).

Context Θ = a learned manifold of regimes (field, subfield, method, venue cluster, citation community, time window, language/register). The metric is *induced by transport inconsistency*, exactly as the paper insists geometry expresses transport structure rather than causing it — so it is learned, not imposed.

Transport, holonomy, curvature = the inductive bias. For a context transition $\theta_a \rightarrow \theta_b$ the model emits a small connection $A_{\theta_a \rightarrow \theta_b}$ (skew-symmetric for an $SO(k)$ fiber, k small ≈ 16 – 32 so matrix exponentials stay cheap and many loops can be sampled). Around a loop, $H_\gamma = \prod \exp(A)$ and $\kappa_\gamma = \|H_\gamma - I\|$. This is HKR’s ontological gauge curvature, *learned from data*. Low $\kappa \Rightarrow$ near-path-independence $\Rightarrow$ the regime where objectivity localises.

3.1 The subtlety the three threads missed: gauge identifiability

Holonomy and curvature are **gauge-dependent**; only gauge-*invariants* (Wilson-loop traces $\text{Tr } H_\gamma/k$, the spectrum of H_γ , $\|H_\gamma - I\|$) are meaningful. A freely-learned connection A has a large redundant symmetry (basis rotations of the fiber), and a naïve network will happily absorb arbitrary rotations into A that inflate or deflate κ without any epistemic content. **Consequence for the architecture:** the learnable object must be regularised toward *flat-as-possible* connections (curvature is paid for, never free), and every reported quantity must be a gauge invariant, not a raw matrix entry. Without this, “ κ predicts stabilisation” can be an artifact of the network’s freedom to choose a basis. None of A/B/C raise this; it belongs in the design.

4 Conceptual architecture (five layers)

The system is a stack of five conceptual layers, each given by *what it is, its HKR correspondence*, and *its specific failure mode*. Implementation (PyTorch Geometric / JAX+diffraction, neighbour sampling, mixed precision) is omitted by intent; that is the build, not the concept.

L1 — Fiber encoder (text $\rightarrow \psi$). *What:* a scientific-text encoder producing a small fiber

vector per claim, fine-tuned with a contrastive objective over relation-typed pairs (same / supporting / contradicting / superseding / unrelated-but-near). *HKR*: the re-identifiable *Faktkern*. *Failure mode*: if ψ encodes topic similarity only, “claim identity” collapses to “same subject” and everything downstream measures topicality. The contrastive relation-typing prevents this.

L2 — Temporal context graph ($\rightarrow O, E, T, R$ proxies). *What*: a dynamic, time-stamped multigraph (citation, co-citation, semantic support/contradiction, method/field transfer) over which time-resolved proxies are estimated. *HKR*: the manifestation field and the fact-state tuple over time. *Failure mode*: leakage — future citations/labels must never enter a node’s pre-cutoff state (Section 5.3).

L3 — Gauge-connection / holonomy core ($\rightarrow \kappa$). *What*: learns A per context transition, composes holonomies on sampled loops, emits gauge-invariant κ . *HKR*: the connection, curvature two-form, Wilson loop — the literal heart of the theory. *Failure mode*: gauge non-identifiability (Section 3.1) and the temptation to let κ correlate with degree/popularity; both handled by invariance regularisation + the confound gauntlet.

L4 — Calibrated differentiable dynamics ($\rightarrow$ trajectories). *What*: a neural ODE/SDE (or, first, a plain discrete recurrence) for $z_i(t) = [\psi, O, E, T, R, \kappa]$, with HKR’s free-energy structure (Eq. 7) entering as a **soft** consistency term. *HKR*: the Langevin / free-energy flow; established vs revision channels gated by κ . *Failure mode* — *the single most important constraint in this note*: the redundancy law $R^{-1/2}$ must **not** be imposed by the loss. Leave the exponent free; estimate it on held-out data; report it from *unregularised* runs too. A baked-in law you then “find” is circular and worthless (Section 6).

L5 — Active-inference / holoeconomic agents (CAPSTONE — exploratory only). *What*: lightweight agents traversing the learned graph, maximising expected future Established $_G$ while paying a holoeconomic cost C_h . *HKR*: §6 cognitive bridge, §7 mimesis, the consensus $\leftrightarrow$ polarisation bifurcation, and the *epistemic analogue* of holoeconomic bending. *Failure mode*: degrees-of-freedom inflation (policy + cost + reward shaping + topology) makes any positive result look engineered. *Rule*: agents run only after L1–L4 produce a confound-surviving result, and any agent finding is reported as *exploratory*, never primary evidence — and explicitly *not* as a test of P6-the-physics-claim.

5 The falsification spine (this is what makes it science)

A model that can only succeed is not evidence. The architecture is organised so that the *first* thing it does is try to kill its own central claim. Three instruments.

5.1 The confound gauntlet (the real Stage 1)

The headline question is never “does κ predict stabilisation” — it is “does κ predict **beyond** the boring explanations”. The HKR model must beat, on identical data/labels/splits, *all* of the controls in Table 4.

If HKR survives both non-negotiable controls, the result is interesting. If it survives neither, you have learned something real and publishable about HKR’s limits — which §13 explicitly frames as still valuable.

5.2 The circularity audit (proxies)

Because O, E, T, R, κ are all distilled from the *same* citation substrate, “discovering” relationships among them risks rediscovering definitions. Guards: *externally-anchored labels* wherever possible (time-to-textbook-absorption, inclusion in independent review literature, cross-domain adoption diversity — quantities not used to *construct* the predictor); *pre-registration* of proxy definitions, success thresholds, the exponent target, and the ablations before the held-out windows are touched (you are the author — this is the cheapest credibility available); and *disjointness*

Baseline / control	The confound it kills
Citation count + paper age	κ is popularity/recency in disguise
PageRank / eigenvector centrality	κ is ordinary network prestige
Text-embedding-only classifier	semantics alone explains stabilisation
Temporal GNN <i>without</i> holonomy (parameter-matched)	ordinary graph learning already suffices
Topic-model / novelty score	“new idea gets popular” explains it
Disruption index (CD / Funk–Owen)	an existing paradigm-shift metric already captures T
Context-label shuffle	the <i>geometry of context</i> is real, not decorative
Degree-preserving edge rewire	structure beyond degree matters
Time-reversal / future-leakage probe	the prediction is genuine, not hindsight

Table 4: The confound gauntlet. The two non-negotiable controls are “same graph, holonomy removed” and “context labels shuffled”.

checks reporting the mutual information between each proxy and the confound baselines (if κ is $\gtrsim 95\%$ explained by citation-count + centrality, it is not a new variable).

5.3 Leakage discipline (temporal)

Strict historical cutoffs: train only on $t_i \leq Y$, predict observable establishment over $Y+1 \dots Y+5$, sweep many Y , multiple domains. Two modes: a *pragmatic* mode (frozen pre-trained embeddings, no future edges) and a *strict* mode (encoders and graph features trained only on pre-cutoff data). The strict mode is slower and is the one that survives review.

6 The redundancy law, done right (κ_{res} vs R)

This deserves its own section because it is load-bearing for HKR and hides the deepest trap — one the careful synthesis only half-saw.

The trap. HKR’s redundancy law is stated as: *the Fréchet mean of holonomies over R independent noisy paths has residual curvature $\kappa_{\text{res}} \propto R^{-1/2}$* . Read literally, if κ_{res} is defined as the residual of an average over R sampled paths, then $\kappa_{\text{res}} \propto R^{-1/2}$ is **just the standard error of the mean** — the central limit theorem — and it appears for *pure noise with zero HKR content*. A pipeline that sets $R =$ “number of paths I averaged” will rediscover the $-1/2$ exponent every time, on any data, including shuffled data. That is not evidence for HKR; it is arithmetic.

The fix (what makes the test non-trivial). The HKR-meaningful version requires two decoupled quantities: (1) κ_{res} estimated at a *fixed* transport sample size for every claim (so the standard-error term is constant across claims and cannot drive the regression); and (2) R as an *external, semantic / sociological* redundancy — the effective number of *independent communities* carrying the claim, measured from a different signal than the one used to estimate κ . Then regress $\log \kappa_{\text{res}}$ on $\log R$ on held-out windows with the exponent **free**:

- Under the **null** (independent communities have nothing to do with transport consistency): $\alpha \approx 0$. The naïve estimator would *still* report $-1/2$ here — which is exactly how you demonstrate the trap and prove your estimator is clean.
- Under **HKR-true** (claims genuinely carried by more independent communities have lower transport inconsistency): $\alpha < 0$, ideally with a 95% CI bracketing $-1/2$, **and** robust to ablation and shuffle.

Only the decoupled, fixed-sample, free-exponent estimate that comes out near $-1/2$ on *external*

redundancy counts toward rung 3. This single correction is the difference between a result a reviewer respects and a result a reviewer deletes in one sentence.

7 Scalability and compute philosophy

The hardware brief — 20 GB VRAM, up to 60 GB RAM, willing to run hours→weeks — is genuinely well-matched, but for a specific reason that must be stated honestly.

More compute buys statistical power, not significance. The thesis’s “support is monotone in resources” is true *only* in the sense that longer runs tighten confidence intervals, sweep more cutoffs, sample more loops, and run more bootstrap/permutation replicates and more seeds/domains. It does **not** manufacture an effect that is not there — and the design makes that impossible (free exponents, held-out windows, pre-registered thresholds). Framed this way, the resource ladder is a *precision* ladder, mirroring the engine’s own philosophy without inheriting its within-model circularity.

Architectural shape that makes this scale. The graph stays on CPU/RAM; the GPU sees mini-batches. Real citation graphs are memory-heavy; 60 GB RAM holds time-sliced shards, memory-mapped **fp16** embeddings, and neighbour-sampling buffers. The 20 GB GPU does only the dense work: small matrix exponentials, loop holonomies, the ODE/SDE solves, contrastive fine-tuning — none of which needs the whole graph resident. A small fiber dimension ($k \approx 16\text{--}32$) keeps holonomy cost low so the expensive axis becomes *number of loops* $\times$ *cutoffs* $\times$ *seeds* — i.e. precision, which is exactly the axis worth spending weeks on. Data is *tiered* rather than “all of science”: start with arXiv-scale metadata (light), graduate to Semantic Scholar / S2ORC for abstracts + citation context, and only then to OpenAlex slices via temporal subgraph sampling. Pick one bounded, revision-rich domain first (ML 2010→ for dense fast paradigm shifts, or quantum-information / superconductivity for closeness to HKR’s own Quantum-Darwinism language).

Budget	Goal	What it produces
Hours	<i>Pipeline sanity</i>	20k–100k papers, one domain, paper-level nodes, first κ estimates; does κ correlate with later adoption <i>at all</i> .
Days	<i>First real evidence</i>	0.5–2M papers, multiple cutoffs, the full confound gauntlet , bootstrap CIs, ablations, shuffle controls. Rungs 1–2.
Weeks	<i>Thesis-grade battery</i>	Multi-domain replication, deep ensembles, many cutoff years, the decoupled $\kappa_{\text{res}}\text{--}R$ estimate, finite-size scaling of revision cascades, context-lability sweep, and the full contested→law-like continuity result with foundational-reversal cases. Rungs 3–5.

Table 5: The compute ladder reframed as a statistical-power ladder.

8 Staged programme with gate conditions

The ordering is C/Gemini’s (validator → dynamics → agents), but with **gates**: you do not pay for a later stage until the earlier one clears its falsification bar. This is what stops the project from becoming an expensive way to confirm a foregone conclusion.

- **Stage 0 — Time-sliced dataset + leakage harness + pre-registration.** Define proxies, thresholds, exponent target, ablations *now*. *Gate*: strict/pragmatic split reproducibly built; no future leakage detectable in a placebo test.
- **Stage 1 — Confound gauntlet on paper-level nodes (no holonomy yet).** *Gate*: an HKR-feature model beats citation-count, centrality, text-only, disruption-index, and a parameter-matched plain temporal GNN, out-of-sample, in ≥ 1 domain. *If it fails here,*

stop and report the negative result — it is real.

- **Stage 2 — Add the gauge / holonomy core.** *Gate:* κ adds out-of-sample value and context-shuffle destroys it and holonomy-ablation (parameter-matched) degrades it. Rung 4.
- **Stage 3 — Calibrated differentiable dynamics.** Report *with and without* the soft free-energy term; the headline is strongest when both agree. *Gate:* the decoupled $\kappa_{\text{res}}-R$ exponent (Section 6) comes out near $-1/2$ on *external* redundancy, robust to ablation.
- **Stage 4 — Claim-family nodes.** Re-run Stages 1–3 with nodes closer to true *Faktkerns*. *Gate:* results survive the unit change (a claim about HKR should not depend on “paper” vs “claim-family”).
- **Stage 5 — Agent capstone (exploratory).** Consensus/polarisation, stuck-paradigm traps, and the *epistemic* analogue of holoeconomic bending. Labelled exploratory; not primary evidence; explicitly not a physics test.

9 The self-application meta-test (kept honest)

HKR’s §11 self-application is elegant and worth mirroring: encode the ML programme’s *own* design (its layers, losses, the claim “this validator is self-consistent”) as a small fact-field and run the dynamics; if it relaxes to low- κ / high-Established_G, that is a satisfying internal consistency check. State plainly what it is and is not: **a coherence property of the formalism, not evidence about science or nature.** It is a nice closing figure, not a rung on the ladder.

10 What success and failure look like

The honest headline (the most you should claim — and it is a lot):

*On real scientific citation histories, a temporal model carrying HKR’s gauge-theoretic structure — learned holonomic curvature, two-channel dynamics, and redundancy — predicts the future stabilisation of claims better than citation-count, prestige, novelty, disruption-index, and ordinary temporal-graph baselines, across multiple domains and many historical cutoffs. Curvature relaxes with **independent** redundancy at an exponent consistent with $-1/2$ under a decoupled estimator; high- κ /high- T claims show greater cross-context lability and distinct revision trajectories; and removing the holonomy geometry measurably degrades prediction while shuffling context destroys the effect. Most distinctively, claims trace **one continuous κ/R relaxation from contested to law-like** — with foundational reversals appearing as the tail of that same process, not a separate category — which is direct resonance for HKR’s unification claim: an effective law as the low- κ / high- R limit of one drifting field. This supports HKR as a predictive theory of epistemic dynamics, in the very regime the theory says comes before the laboratory. It tests the **epistemic mirror** — how claims stabilise and drift within human knowledge — and by design says nothing about whether the **physical referents** themselves drift (P5/P6 in the world), which remain for the §13 programme.*

Failure — and why failure is also a contribution. HKR is counted *against* if: learned κ adds no out-of-sample value; κ is $\gtrsim 95\%$ explained by popularity/centrality; the decoupled $\kappa_{\text{res}}-R$ exponent is null while only the confounded one shows $-1/2$; shuffled context performs equally; high- T /high- κ claims show no distinct dynamics; the effect lives in one domain and dies elsewhere; or the soft free-energy term improves aesthetics but not prediction. Any of these tells you *which* part of HKR needs revision — which, by the Revisionsaxiom itself, is exactly how a high- T framework earns its eventual establishment rather than being protected from test.

11 Kill-criteria and redesign triggers

The conditions under which this *specific* programme should be abandoned or redesigned rather than pushed harder: (i) the proxies cannot be made disjoint from the confound baselines (everything is popularity) $\rightarrow$ redesign the labels around purely external signals; (ii) gauge non-identifiability cannot be controlled and κ tracks the network’s basis freedom $\rightarrow$ restrict to Wilson-loop invariants only, or drop the learned connection for a fixed-transport baseline; (iii) the redundancy law only ever appears in its confounded (CLT) form $\rightarrow$ report it as a null for HKR-specific content; (iv) results are domain-bound $\rightarrow$ reframe as a domain-specific phenomenon, not support for a universal field theory. None of these are embarrassments; each is the kind of bounded, honest finding that makes the surrounding positive results believable.

Appendix Mapping HKR objects to ML constructs

HKR object	ML construct	Reported as
Fact-state $x_i = (O, E, T, M(\theta))$	Node feature tuple over time	—
Fiber ψ_i / <i>Faktkern</i>	Contrastively fine-tuned claim embedding	identity-stable across contexts
Context point/space θ, Θ	Learned context manifold (field/method/community/time)	—
Connection A , holonomy H_γ	Learned skew-symmetric transport + loop product	gauge invariants only
Curvature $\kappa = \ \Omega\ $, $\kappa_\gamma = \ H_\gamma - I\ $	Norm of learned curvature / loop deviation	invariant scalar
Redundancy R_δ	External effective # independent communities	decoupled from κ sample size
Free energy F (Eq. 7)	Soft consistency term, not a hard constraint	with/without ablation
Fisher–Rao g^{FR}	Natural-gradient update (efficiency test, P3)	optional
Mutable core / ontology history (C15)	Consensus / status-reversal case studies	exploratory, observable proxy
Context-lability s_i (C16)	Cross-context variance of O vs learned κ	rung 3
Active inference / mimesis / C_h (§6–§7, P6)	Agent capstone	exploratory; epistemic analogue, not physics